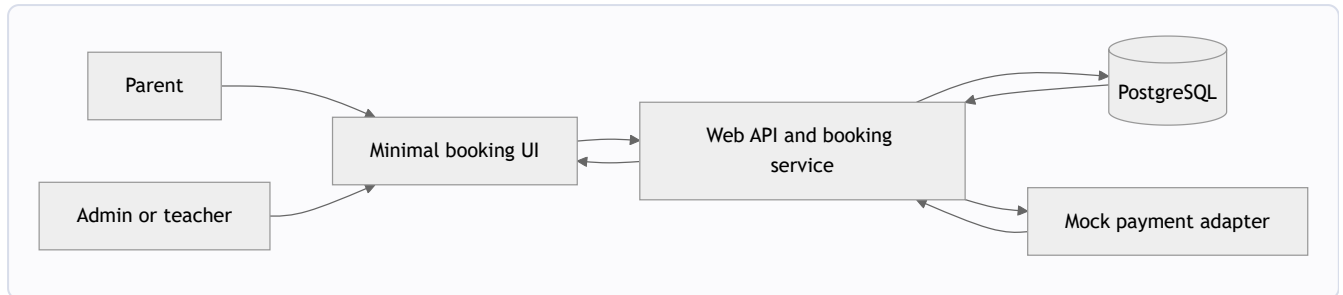


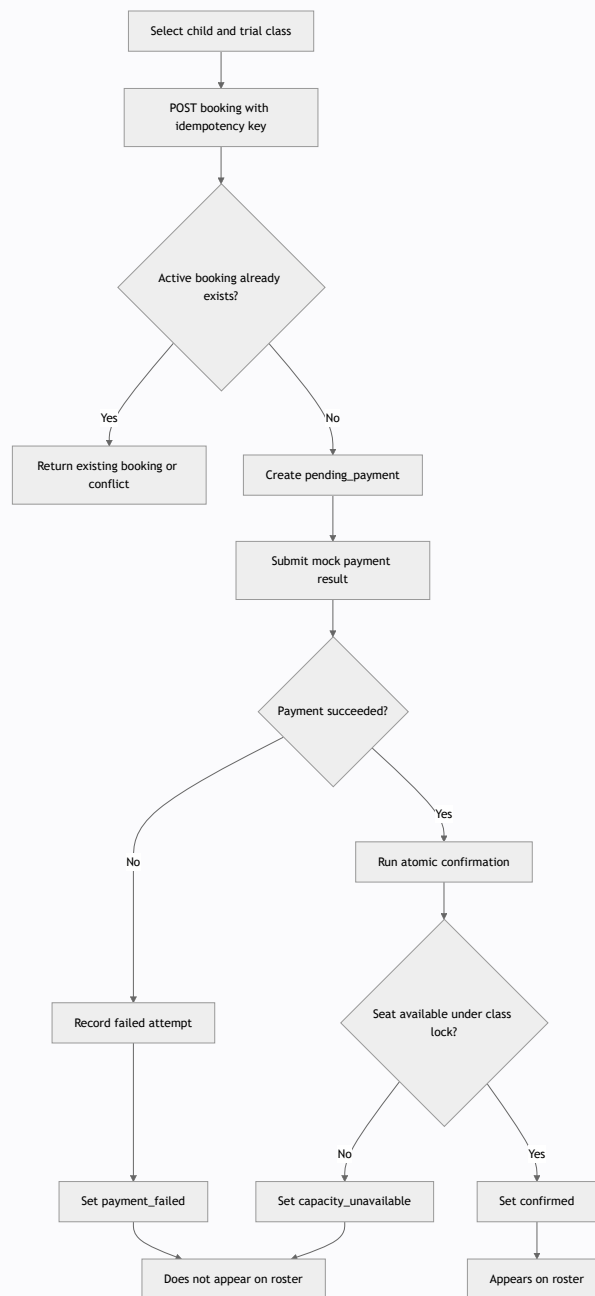
Flowcharts

These Mermaid diagrams are documentation artifacts; they do not prescribe a particular framework.

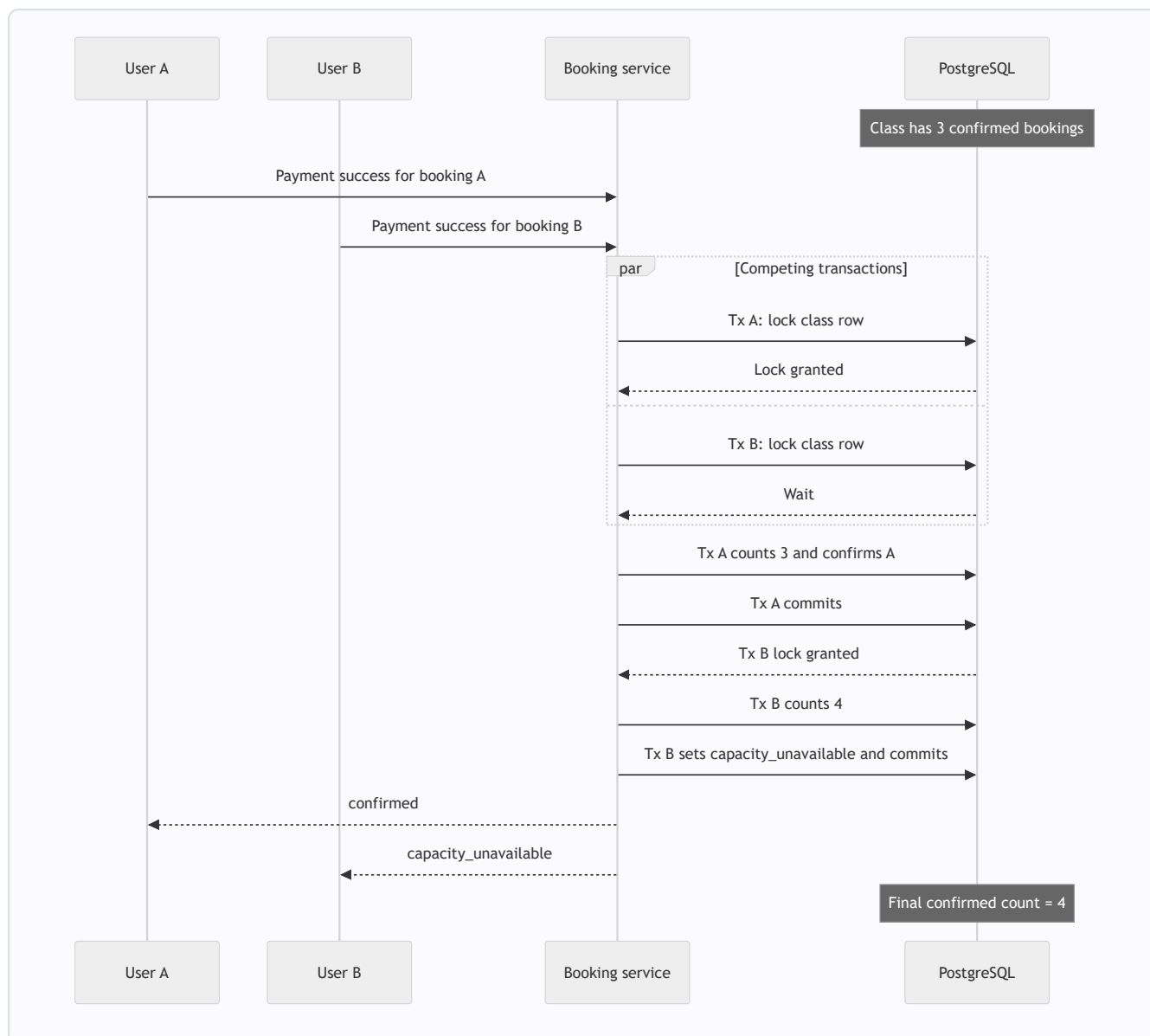
System context



Happy path and payment failure

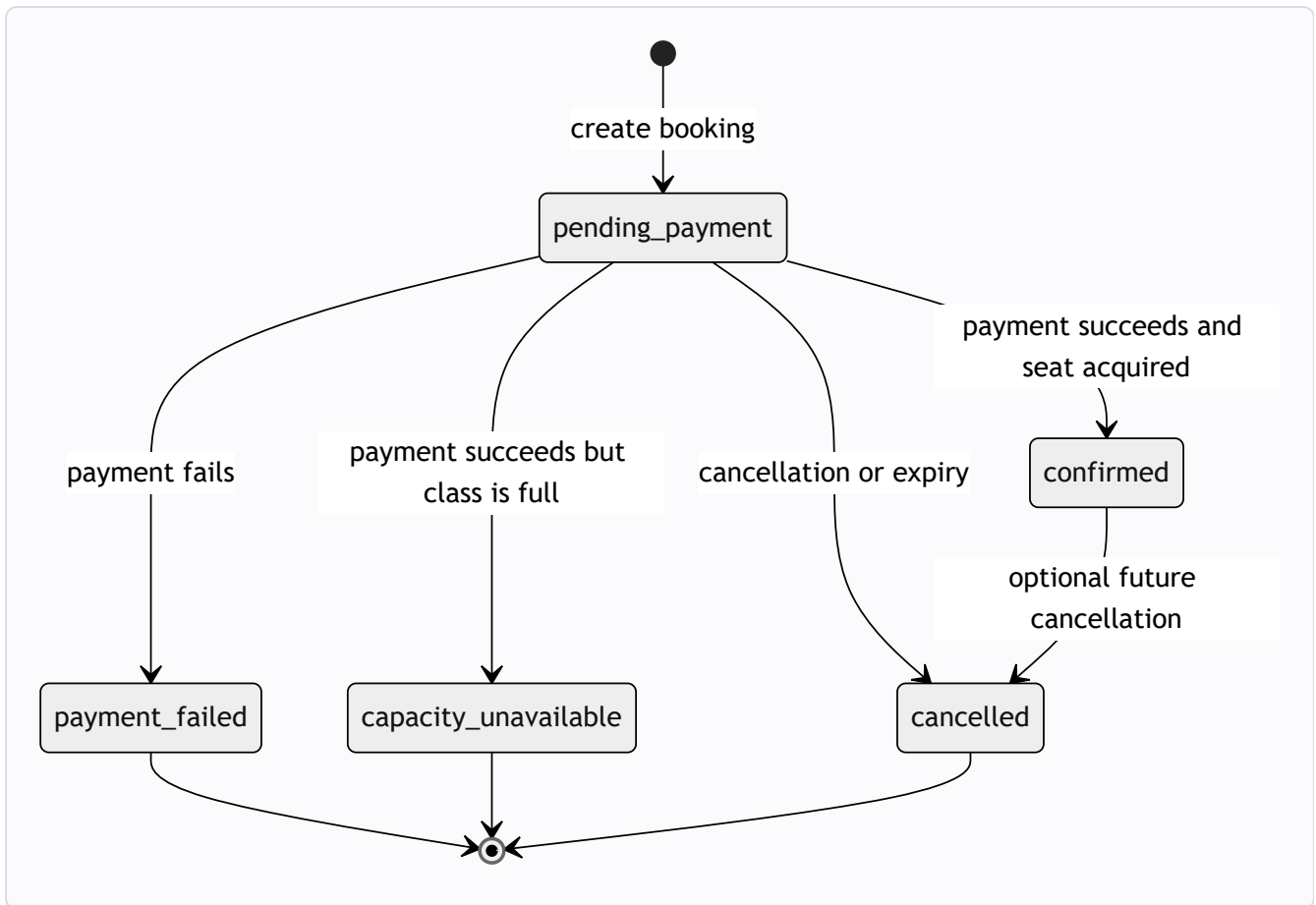


Last-seat race sequence



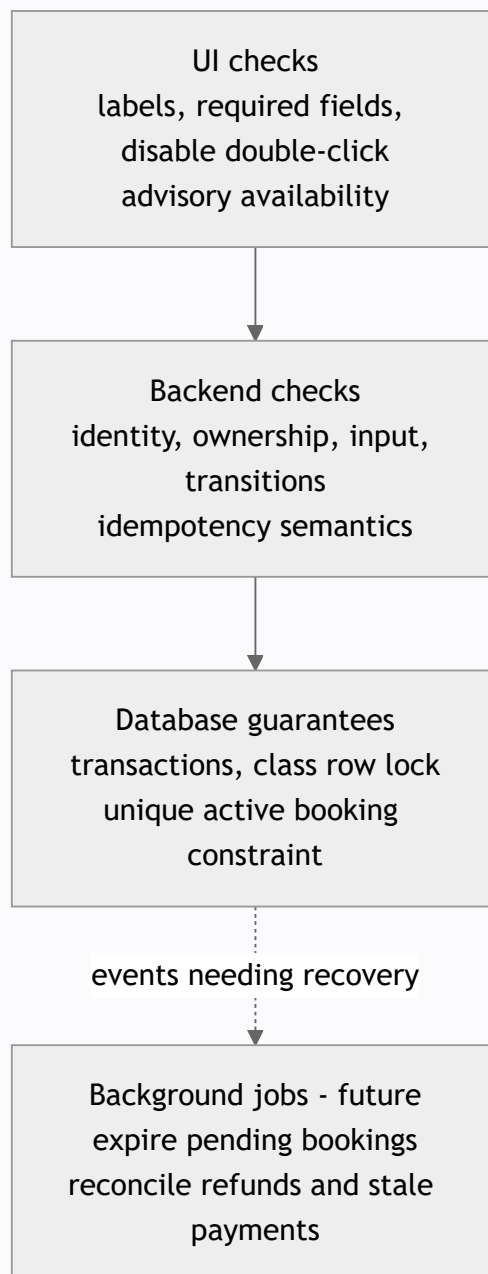
The winner may be A or B. The invariant is that exactly one wins.

Booking state machine

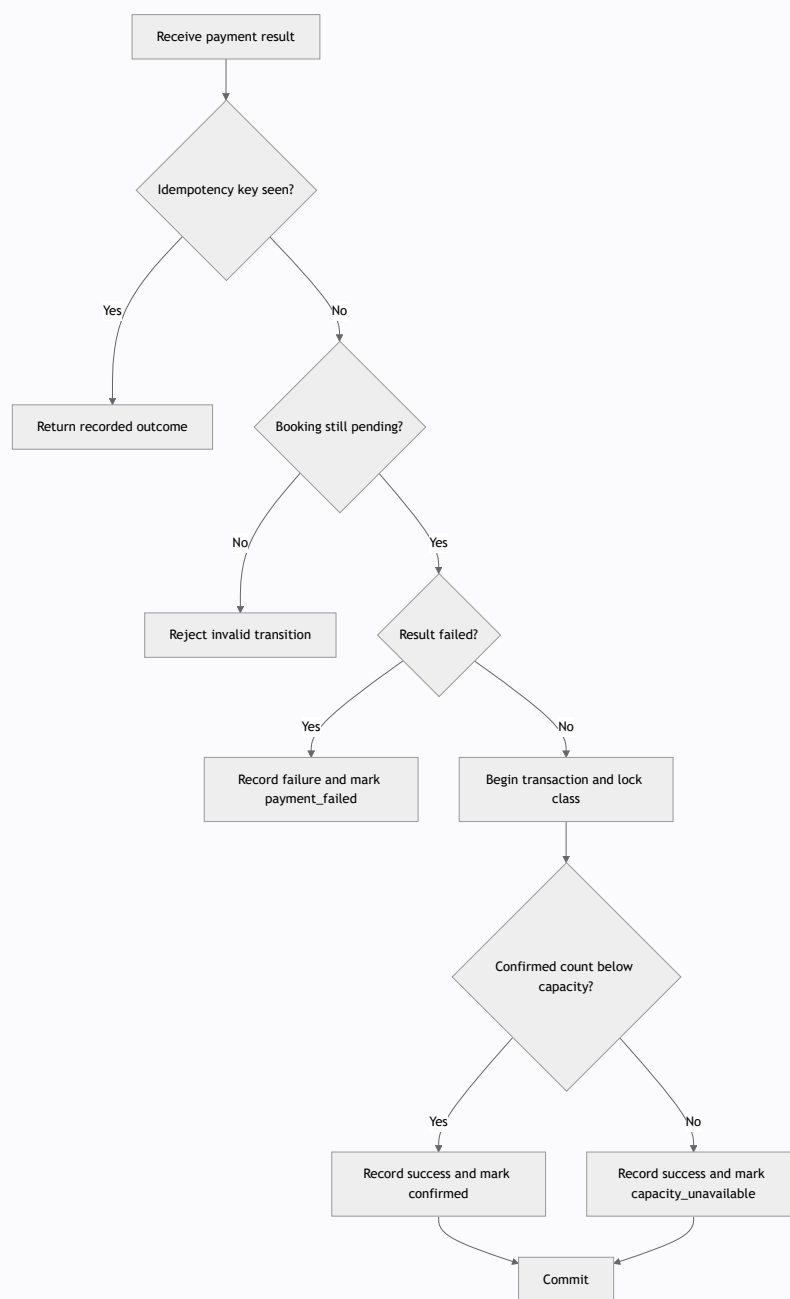


For the demo, terminal states reject contradictory later payment results. A new booking may be created after a failed, unavailable, or cancelled booking.

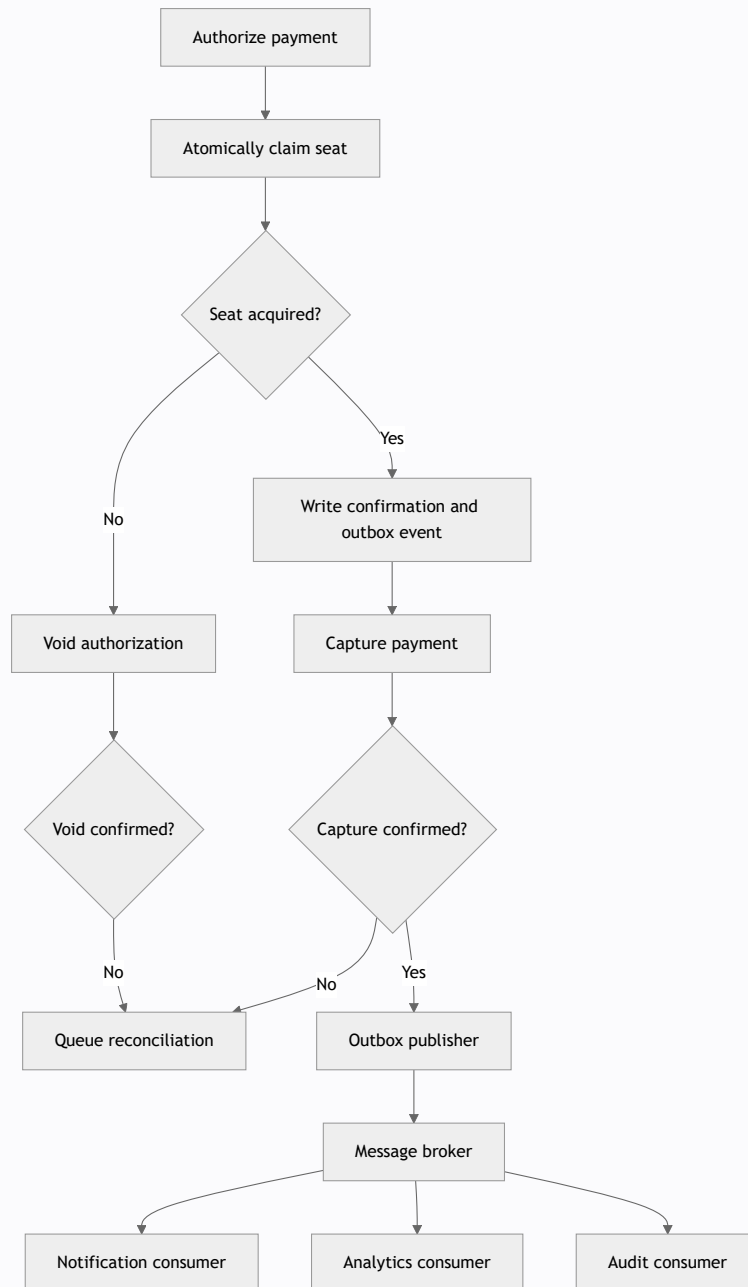
Authority by layer



Payment-to-seat decision



Production evolution: Saga-like compensation and optional event streaming



The implemented take-home stops at the synchronous database-backed flow shown earlier. This production diagram is intentionally future-facing. The broker is not assumed to be Kafka; Kafka becomes a candidate only when replay, throughput, ordering, retention, and independent-consumer requirements justify it.